

Characteristic Geometry and Portfolio Choice

From return forecasts to stock holdings

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When do similar forecasts imply different portfolios?

Value and profitability signals can emphasize many of the same stocks. More signals need not give an investor an equally broad set of exposures.

A factor model describes returns. An investment rule must also decide **which fitted means to use** and **which exposures to penalize**.

We separate two questions:

1. Can changing the units of factor loadings change the risk penalty, even when stock-return forecasts barely change?
2. Within one fitted model, how much does using its retained intercept change portfolio performance?

The object is the conversion of a fitted return model into holdings.

Rolling U.S. equity evidence, with 132 characteristics. The first question concerns the investment rule's risk input; the second concerns its mean input.

From IPCA estimation to portfolio-input comparisons

Kelly, Pruitt, and Su (2019). Instrumented Principal Components Analysis (IPCA) uses characteristics to model time-varying factor loadings and investigates anomaly intercepts.

Our complementary question: what changes when these estimates are used to choose stock holdings?

Comparison	What is varied?	What can it establish?
Risk input	Loading units; then remove only the factor-covariance ridge	Source of a holdings gap
Mean input	Use the intercept from one unchanged model fit	Source of a gross-return increment

Separate new forecast information from a different use of existing estimates.

A ridge is a diagonal covariance adjustment. The companion theory explains metric-indexed attribution; the empirical comparisons here examine portfolio inputs. Neither comparison is a claim to replace IPCA. References: Kelly et al. (2019); Liu (2026), unpublished theory paper.

Model setup: factor terms and a retained mean

Within each estimation window, stock i 's next-month excess return is

$$r_{i,s+1} = z'_{i,s} \Gamma_{\beta} f_{s+1} + z'_{i,s} \Gamma_{\delta} g_{s+1} + \underbrace{z'_{i,s} \Gamma_{\alpha}}_{\text{retained intercept}} + \varepsilon_{i,s+1}.$$

$z_{i,s}$: 132 characteristics; f : three latent factors; g : five observed Fama–French factors. The Γ matrices map characteristics into loadings; ε is the stock-specific disturbance.

The metric specifies how loading lengths and orthogonality are measured.

Fitted specification	Loading metric	Intercept
IPCA comparator	Euclidean, I_L	Set to zero
QZ-IPCA	Characteristic metric, Q	Set to zero
NA-IPCA	Same characteristic metric	Exposure budget

NA-IPCA: No-Arbitrage IPCA; the name motivates control of retained mean exposure (appendix: pricing motivation). These are this paper's fitted comparators. Q : characteristic metric; I_L : identity; $L = 132$.

The Gram matrix measures stock exposures

Let Z_s stack the characteristics of N_s stocks at date s . For two characteristic-loading directions $u, v \in \mathbb{R}^L$,

$$W_s = \frac{Z_s' Z_s}{N_s}, \quad u' W_s v = \frac{1}{N_s} \sum_{i=1}^{N_s} (z'_{i,s} u)(z'_{i,s} v).$$

- Setting $u = v$ gives average squared stock exposure: $v' W_s v = \|Z_s v\|^2 / N_s \geq 0$.
- Diagonal entries record characteristic second moments; off-diagonal entries record their cross-products.
- $u' v = 0$ separates coefficient directions. It need not separate the stock exposures they generate.

The empirical ruler compares induced stock exposures, not just coefficients.

W_s is symmetric and positive semidefinite, with rank equal to that of Z_s . It is a second-moment matrix, not generally a demeaned covariance matrix. Rank scaling individual characteristics does not remove their overlap.

Changing the ruler: IPCA, QZ-IPCA, and NA-IPCA

IPCA uses the coefficient-space ruler I_L . QZ-IPCA updates its ruler from trailing stock-exposure geometry:

$$\hat{Q}_t = \sum_{s \in \mathcal{W}_t} a_{s,t} W_s, \quad a_{s,t} \geq 0, \quad \sum_{s \in \mathcal{W}_t} a_{s,t} = 1.$$

Here $\mathcal{W}_t$ is the training window and $a_{s,t}$ its declared weights. For either declared ruler Q , normalization and separation use

$$\Gamma'_\beta Q \Gamma_\beta = I_K, \quad \Gamma'_\beta Q [\Gamma_\delta, \Gamma_\alpha] = 0.$$

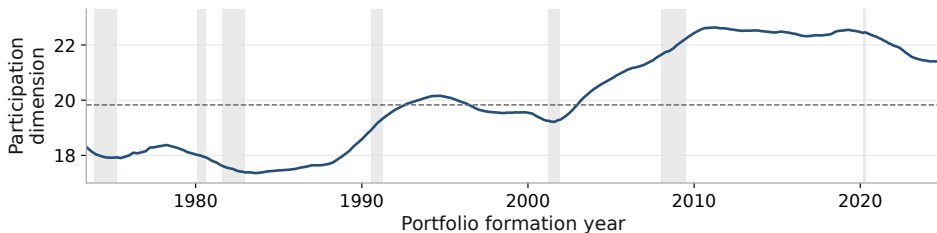
NA-IPCA additionally bounds retained intercept exposure, $\Gamma'_\alpha Q \Gamma_\alpha \leq \delta_\alpha$. The budget δ_α is an extra restriction, not a ruler choice.

The ruler sets loading units; the portfolio rule determines how those units enter a risk penalty.

$Q_Z = \mathbb{E}(W_s)$ is the population Gram metric; $\hat{Q}_t$ is its rolling empirical counterpart. Positive definiteness requires support in every direction across the weighted window; all saved metrics are positive definite. $K = 3$. Exposure orthogonality is distinct from zero return covariance.

132 characteristics, about 20 effective directions

How different is the empirical ruler from equally dispersed characteristic variation? Examine the saved rolling metrics.



Median spectral participation dimension: **19.8**. Characteristic variation is concentrated despite the large signal count.

Participation dimension: $(\text{tr } Q)^2 / \text{tr}(Q^2)$; it equals 132 for equal eigenvalues. The detail axis starts at 17; all 132 directions remain supported. 618 exponentially weighted metrics, fixed characteristic count. Gray: NBER recessions at formation dates. Rolling preprocessing also affects the path.

A common portfolio formula uses both mean and risk

Core U.S. stocks excluding micro-caps; 120-month estimation windows. At each date t , form holdings for the return at $t + 1$.

The factor-only forecast and regularized stock-risk matrix are

$$\hat{\mu}_t = B_t \hat{m}_{F,t}, \quad A_t = B_t (\hat{C}_{F,t} + \lambda_F I) B_t' + \hat{D}_t + \lambda_\Sigma I_{N_t}.$$

$B_t = Z_t [\hat{\Gamma}_{\beta,t}, \hat{\Gamma}_{\delta,t}]$: stock loadings on all eight factors; $\hat{m}_{F,t}$ and $\hat{C}_{F,t}$: training factor mean and covariance; $\hat{D}_t$: diagonal residual risk. N_t : stock count. Factor ridge λ_F and stock ridge λ_Σ both equal 0.01.

$$\tilde{w}_t = A_t^{-1} \hat{\mu}_t, \quad w_t = \frac{\tilde{w}_t}{\sum_i |\tilde{w}_{i,t}|}.$$

The mean rewards a position; inverse risk adjusts for its variance and overlap with other positions. Unit-gross scaling fixes total absolute holdings.

A change in B_t can alter the penalty $\lambda_F B_t B_t'$ even when forecasts agree.

Here w_t is the stock-weight vector and $\tilde{w}_t$ its unscaled mean–variance solution; I is an identity matrix. Same universe and formation dates within each comparison. The metric comparison uses 618 months (June 1973–November 2024); the later same-fit comparison uses 594. Unit gross does not impose dollar neutrality or equal volatility.

Forecasts nearly agree; stock holdings do not

First comparison: IPCA versus QZ-IPCA, with the same portfolio routine. The characteristic metric weights the trailing 120 months exponentially.

Diagnostic across 618 months	Median	What it tells us
Relative forecast gap	4.7×10^{-6}	Mean inputs nearly agree
Relative residual-variance gap	4.2×10^{-7}	Residual risk nearly agrees
Weight cosine	0.830	Holdings point in different directions
Relative raw-weight gap	0.607	Unscaled holdings differ

What in the risk input can account for this holdings gap?

Cosine 1 denotes identical weight directions; relative gaps use IPCA as denominator. Forecast agreement compares two model forecasts, not forecasts with realized returns. EWMA means exponentially weighted moving average; 120 months, 60-month half-life. Source: archived paper mechanism diagnostics.

A fixed factor ridge changes the exposure penalty

In a mean–variance objective, the factor ridge adds the penalty

$$\frac{\lambda_F}{2} w' B B' w = \frac{\lambda_F}{2} \|B' w\|^2.$$

Here $B'w$ is the portfolio's vector of factor exposures. The equality uses $w' B B' w = (B' w)' (B' w)$.

Consider a change of units for one factor:

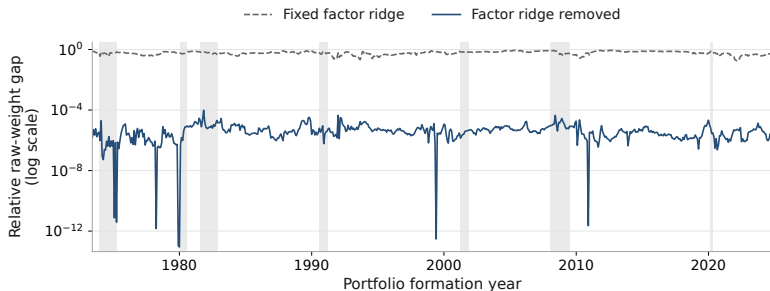
- Double the factor and halve its stock loadings. Consistently transformed means and covariances give the same stock forecast and unregularized stock risk.
- Keep the numerical ridge unchanged. The penalty on that loading direction falls to **one quarter**.

A numerical stabilization device has become an exposure preference.

Algebraic illustration, not another performance test. Factor normalization sets loading units; holding the factor ridge fixed need not preserve the same investment rule. The appendix gives the general change-of-coordinates identity and the adjustment that preserves stock risk.

Removing only the factor ridge closes the holdings gap

Keep forecasts, residual risk, and the stock ridge fixed. Remove the factor ridge and re-solve the saved portfolio systems.



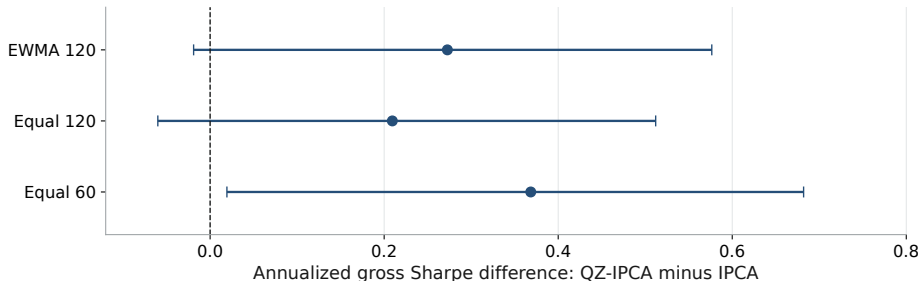
Median relative raw-weight gap: **0.607** before removal; 3.9×10^{-6} after removal.

The controlled re-solve identifies the ridge's role in holdings. The next question is whether the original portfolios deliver a reliable return advantage.

618 matched windows. Maximum remaining gap: 9.6×10^{-5} . Log scale; all positive observations retained. Gray: NBER recessions. This comparison fixes the model estimates; it does not evaluate realized returns on ridge-free portfolios.

A holdings difference is not yet a reliable return gain

QZ-IPCA minus IPCA: annualized gross Sharpe differences and 95% intervals.



All three point estimates are positive, but two intervals include zero. The equal-60 interval also includes zero with longer resampling blocks (appendix).

Risk-input evidence is clear; the performance ranking is less precise.

618 paired months. EWMA 120: exponential weights, 60-month half-life. Equal 120 and equal 60: uniform weights over those trailing months. All three recorded rules are reported. Paired 12-month moving-block bootstrap, 10,000 draws; pointwise intervals, not selection-adjusted. Unit-gross portfolios, gross of costs.

Next, change the mean input within one NA-IPCA fit

The first comparison concerned a risk penalty. Now hold one NA-IPCA fit fixed and ask what the investor gains from **using its retained intercept**.

$$\hat{\alpha}_t = Z_t \hat{\Gamma}_{\alpha,t}, \quad \tilde{w}_t^{\text{mean}} - \tilde{w}_t^{\text{off}} = A_t^{-1} \hat{\alpha}_t.$$

$\hat{\alpha}_t$ is the stock-level intercept forecast. Holding A_t fixed, it changes unscaled holdings through inverse stock risk.

Portfolio rule	Mean input	Additional risk input
Off	Factor mean	None
Mean-only	Factor mean + intercept	None
Complete	Factor mean + intercept	Loading instability

Off is not a re-estimated QZ-IPCA model. Only investment inputs change.

594 common months: June 1975–November 2024, after 24 prior loading estimates. EWMA metric; one NA-IPCA fit per window. The budget $\Gamma'_\alpha Q \Gamma_\alpha \leq 0.035$ is slack in all saved windows. Complete adds risk based on variation in 24 prior intercept-loading estimates (multiplier one). Each rule is separately scaled to unit gross.

The mean input explains most of the gross increment

Gross performance	Off	Mean-only	Complete
Annualized mean (%)	1.75	7.59	7.60
Annualized volatility (%)	4.46	3.02	3.00
Annualized Sharpe ratio	0.3917	2.5126	2.5347

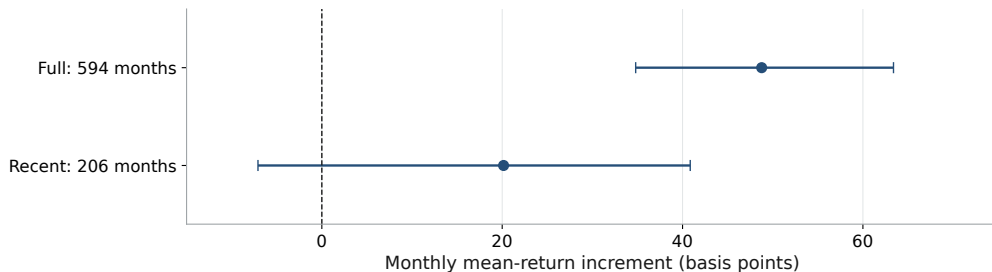
Paired increment	Estimate	95% interval
Mean-only minus off: monthly mean (bp)	48.69	[34.66, 63.36]
Complete minus mean-only: monthly mean (bp)	0.08	[-0.15, 0.26]
Complete minus mean-only: Sharpe	0.0222	[0.0100, 0.0355]

The dominant change is the mean input. The instability adjustment adds a small Sharpe increment in this full sample.

One fitted model; 594 common months; unit gross. Paired 12-month moving-block intervals, 10,000 draws; bp: basis points. Loading instability is a risk proxy. Source: archived input-ablation table and inference outputs.

The recent-sample increment is less precisely estimated

Complete minus off: monthly mean-return difference. The recent period is a subset of the full sample.



The recent point estimate remains positive, but its interval includes zero. The historical full-sample gain is not a precise estimate of the recent opportunity.

Separate the mechanism from the size and persistence of its payoff.

Full sample: June 1975–November 2024, 594 formation months. Recent: October 2007–November 2024, 206 months. Paired pointwise 95% intervals; these are not a formal test of a difference between periods. Unit-gross returns, gross of costs.

What should change in portfolio evaluation?

1. **Make the exposure penalty explicit.** For a pure change of factor units, transport the covariance adjustment with the factors, or use the same stock-space adjustment.
2. **State which fitted mean the investor uses.** Compare mean-only and complete rules within one fit before attributing performance to a new model or to a risk correction.
3. **Evaluate the resulting investment rule.** Use paired dates, report all recorded metric rules, and examine sample and dependence sensitivity alongside gross performance.

Geometry makes a hidden portfolio choice observable and testable.

The matched-change result preserves stock forecasts and unregularized stock risk; it does not select an optimal metric. A deployment evaluation would additionally require ridge-free return paths, verified information timing, and trading costs based on actual holdings.

Conclusion: distinguish estimation from investment inputs

Why can similar forecasts produce different portfolios? A fixed factor ridge penalizes exposure in normalization-dependent units. Removing it nearly eliminates the holdings gap in the matched comparison.

What explains the same-fit gross-performance increment? Restoring the intercept mean accounts for most of it. The additional instability adjustment is much smaller.

What does the paper add? An empirical account of how characteristic-based estimates enter portfolio choice, complementing IPCA's return-model framework.

Compare forecast information, exposure penalties, and mean usage separately.

The risk-input and mean-input comparisons use different matched samples and answer different questions. The contribution is a controlled explanation of recorded portfolio differences, not a universal performance ranking.

Evidence appendix: questions and supporting exhibits

Where can each part of the argument be checked?

Question	Evidence
Why change the ruler?	Model restrictions; exposure geometry
Why do holdings differ?	Diagnostics; removal and initialization checks
Do returns improve?	All metric rules; samples and block lengths
Which intercept input matters?	Full ablation; universes and rolling windows
How do strategies compare?	Forecasts, gross scaling, paths and cost proxies

The appendix preserves every table and figure used in the paper.

Scope: V8 empirical manuscript and appendices, 15 tables and six figures. Long tables and multi-panel figures are split for readability. No additional model estimation.

Model restrictions and empirical comparisons

Which restriction changes across fitted specifications?

Specification	Loading restrictions	Empirical question
IPCA	$Q = I_L, \Gamma_\alpha = 0$	Euclidean reference for normalization and portfolios.
QZ-IPCA	$Q = \hat{Q}_t, \Gamma_\alpha = 0$	Does a metric change alter the portfolio rule when forecasts remain nearly fixed?
NA-IPCA	$Q = \hat{Q}_t, \Gamma'_\alpha Q \Gamma_\alpha \leq 0.035$	What changes when the portfolio retains the fitted intercept in its mean and adjusts for its instability?

QZ-IPCA changes the ruler; NA-IPCA additionally retains a bounded intercept.

Three latent and five observed factors; 120-month windows. I_L : identity; $\hat{Q}_t$: trailing Gram metric; Γ_α : intercept loading. The budget is 0.035 and is slack in all saved windows. IAB denotes the intercept-aware portfolio rule.

Evidence map: geometry and the risk input

What is varied, what stays fixed, and where is the result?

Question	Comparison and evidence
How dispersed and stable are the measured exposures?	All 618 saved EWMA Gram matrices; the feature count remains 132. Evidence: A7; A6
Why can nearly equal forecasts give different weights?	EWMA QZ-IPCA versus IPCA; 618 windows. Remove only the factor ridge from saved portfolio inputs. Evidence: A10; A11
Does a different metric improve realized performance?	Each of three QZ-IPCA rules versus IPCA, on the same months; 618 and 594 months reported separately. Evidence: A14, A15, and A18

Each comparison answers a different question; its held-fixed inputs and sample matter.

Available history: 618 formation months, June 1973–November 2024. Strict-24: 594, June 1975–November 2024, after 24 prior loading estimates. References point to appendix slides.

Evidence map: mean inputs and full strategies

What is varied, what stays fixed, and where is the result?

Question	Comparison and evidence
Which intercept input carries the gain?	One EWMA NA-IPCA fit; omit both inputs, restore the mean, then add instability risk; 594 months. Evidence: A21 and A23
How do full strategies compare?	Separately fitted strategies on 594 common months; all three QZ-IPCA rules shown. Evidence: A27 ; A28

Each comparison answers a different question; its held-fixed inputs and sample matter.

Available history: 618 formation months, June 1973–November 2024. Strict-24: 594, June 1975–November 2024, after 24 prior loading estimates. References point to appendix slides.

Evidence map: forecasts and robustness

What is varied, what stays fixed, and where is the result?

Question	Comparison and evidence
Do forecast rankings explain portfolio rankings?	Stock-level forecast metrics on the 594-month common sample; QZ-IPCA uses rolling 60. Evidence: A20 ; A19
How sensitive are the conclusions?	Common initialization; 12/24/60-month resampling blocks; instability multiplier and turnover-proxy checks. Evidence: A12 , A19 , and A26 ; A30

Each comparison answers a different question; its held-fixed inputs and sample matter.

Available history: 618 formation months, June 1973–November 2024. Strict-24: 594, June 1975–November 2024, after 24 prior loading estimates. References point to appendix slides.

Concentration is substantial, but not constant

How concentrated and how stable is the measured characteristic space?

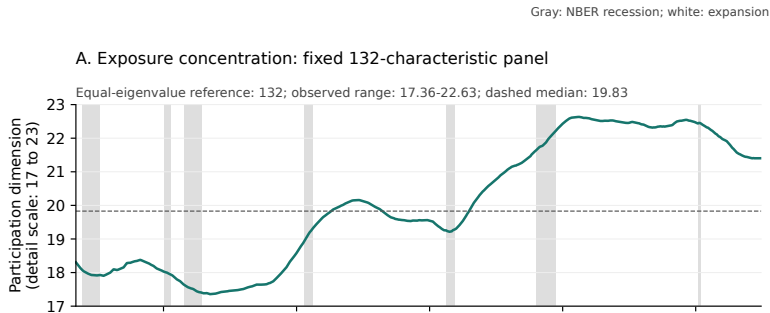
Diagnostic	Minimum	Median	Maximum
Participation dimension	17.36	19.83	22.63
Largest eigenvalue share (%)	11.76	14.24	15.21
Spectral drift (percentage points)	0.033	0.063	0.115
Condition number	1,735	5,922	13,747

The effective dimension is near 20, while the metric's shape and conditioning vary.

618 EWMA metrics, fixed 132-characteristic panel; spectral drift has 617 observations. Participation: $(\text{tr } Q)^2 / \text{tr}(Q^2)$; condition number: largest over smallest eigenvalue. These are exposure-geometry diagnostics, not counts of priced risks.

Effective exposure dimension over time

Is the fixed set of 132 characteristics equally dispersed?

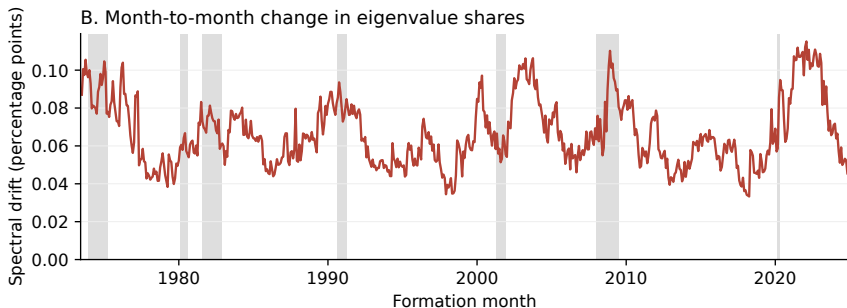


No: the participation dimension ranges from 17.36 to 22.63.

Original panel A; detail axis, not a zero baseline. Formation dates: June 1973–November 2024, shared with panel B. Gray: NBER recessions. Rolling preprocessing also contributes to the path.

Changes in the spectrum reveal changing geometry

Can geometry move even when its effective dimension looks stable?

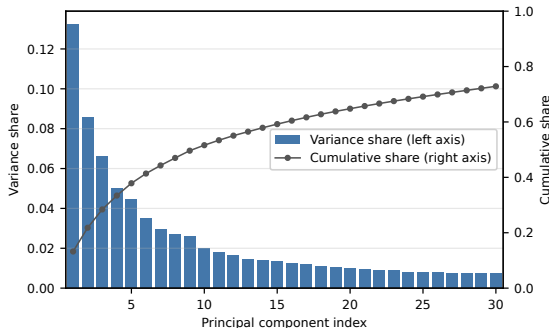


Adjacent-window spectral shares change; dimension alone does not describe that movement.

Original panel B. Spectral drift: $\frac{1}{2} \sum_j |p_{j,t} - p_{j,t-1}|$, with $p_{j,t} = \lambda_{j,t} / \sum_\ell \lambda_{\ell,t}$; axis is in percentage points. 617 changes; gray: NBER recessions. This is spectral change, not eigenvector rotation.

A separate public panel also shows concentration

Is overlapping characteristic variation peculiar to the production panel?



A larger public panel supplies a separate descriptive illustration of concentration.

153 characteristics, 768 months (1962–2025), 36,621 firms; not the 132-characteristic experiment. Rank-scaled features; missing values and thin monthly cells are centered. Bars: first 30 eigenvalues; line: cumulative share of the Gram trace, not return variance.

Full diagnostics locate the holdings discrepancy

Do the mean, residual risk, or factor-ridge inputs account for the gap?

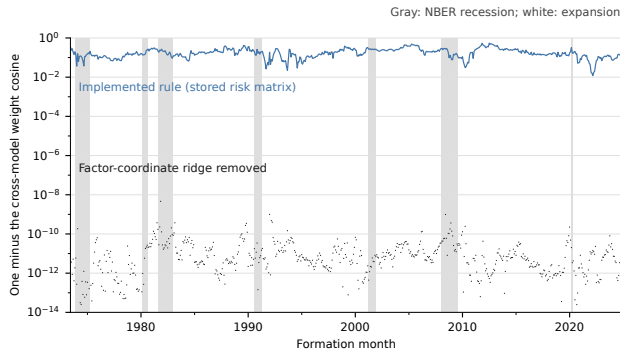
Diagnostic	Median	Maximum or minimum
Mean-term norm / total weight-gap norm (%)	0.07	—
Relative idiosyncratic-variance gap	4.2×10^{-7}	—
Largest latent-span angle (degrees)	4.3×10^{-4}	Max. 0.011
Implemented weight cosine	0.830	—
Relative raw-weight gap before removal	0.607	Max. 0.895
1-cosine after ridge removal	6.0×10^{-12}	Max. 4.6×10^{-9}
Relative raw-weight gap after removal	3.9×10^{-6}	Max. 9.6×10^{-5}
Rebuilt covariance relative gap, both models	—	$< 2 \times 10^{-15}$

Nearly aligned inputs coexist with different weights; ridge removal closes the gap.

618 EWMA windows. Covariance check: 1,236 model-window matrices, factors paired by return date. Norm ratios are not additive return contributions. Cosines describe directions; raw gaps also reflect scale.

Weight directions agree after ridge removal

Does the mechanism remain when leverage scale is removed?



The directional discrepancy nearly disappears, as does the raw-weight gap in the main talk.

618 EWMA windows. Figure plots $1 - \cos(w^{IPCA}, w^{QZ})$, not relative raw-weight gap. Median after removal: 6.0×10^{-12} . Positive rescaling leaves cosines unchanged. Log scale; gray: NBER recessions.

A common initialization retains the mechanism

Could different starting values account for the near-equivalence?

Diagnostic	Median	Maximum
Largest latent-span angle (degrees)	4.3×10^{-4}	1.1×10^{-2}
Relative forecast gap	4.7×10^{-6}	1.4×10^{-4}
Weight cosine, factor ridge retained	0.830	0.988
1-cosine, factor ridge removed	5.8×10^{-12}	4.6×10^{-9}

The same pattern appears when both models use a common SVD initialization.

Separate 618-window check: zero block-loading penalty, moment ridge 10^{-8} , tolerance 10^{-6} , maximum 1,000 iterations; common IPCA residual risk. 617 stopping checks pass; June 1999 reaches the cap in both models and is retained. This is a numerical sensitivity check.

Transporting the adjustment preserves stock risk

What covariance adjustment respects a pure change of factor units?

For an invertible C , write $f^* = Cf$ and $B^* = BC^{-1}$. Transport factor covariance and mean consistently:

$$C_F^* = CC_F C', \quad m_F^* = C m_F.$$

Then $B^* m_F^* = B m_F$ and $B^* C_F^* B'^* = B C_F B'$.

A factor-risk adjustment R_F should likewise become $R_F^* = C R_F C'$:

$$B^* R_F^* B'^* = B R_F B'.$$

Using the same scalar ridge $R_F^* = \lambda_F I$ generally does not do this.

Transport the exposure penalty, not just the forecast and covariance.

Algebraic identity for matched representations. Ridge removal checks the holdings mechanism, not the realized performance of a new ridge-free strategy.

All three metric rules: descriptive performance

How do the recorded QZ-IPCA portfolios compare with IPCA?

Statistic	IPCA	QZ-IPCA		
		EWMA	Equal 120	Rolling 60
Gross Sharpe ratio	0.64	0.91	0.85	1.00
Annualized mean (%)	3.9	4.6	4.5	4.8
Annualized volatility (%)	6.1	5.1	5.3	4.8
Maximum drawdown (%)	-34.6	-18.0	-19.8	-16.5

Higher gross Sharpe coincides with lower volatility and drawdown in these samples.

618 Core months; unit gross, gross of costs. EWMA: 120 months with 60-month half-life; equal 120 and rolling 60: uniform weights. Paired uncertainty, shown next, is needed to interpret the point estimates.

EWMA: full paired inference

Which parts of the performance difference are precisely estimated?

Difference	Estimate	95% interval
EWMA, 120 months		
Annualized gross Sharpe	0.27	$[-0.02, 0.58]$
Annualized mean (percentage points)	0.76	$[-0.45, 2.12]$
Annualized volatility (percentage points)	-1.00	$[-2.32, 0.18]$
Recentered mean-test $p = 0.25$		

The positive Sharpe estimate is not separated from zero at this confidence level.

QZ-IPCA minus IPCA; 618 paired unit-gross months. 95% pointwise percentile intervals, 12-month moving blocks, 10,000 draws. Separately recentered mean test. No adjustment for choosing a metric.

Equal 120: full paired inference

Which parts of the performance difference are precisely estimated?

Difference	Estimate	95% interval
Equal 120		
Annualized gross Sharpe	0.21	$[-0.06, 0.51]$
Annualized mean (percentage points)	0.64	$[-0.54, 1.96]$
Annualized volatility (percentage points)	-0.76	$[-2.06, 0.37]$
Recentered mean-test $p = 0.32$		

The estimated volatility reduction and return gain remain imprecise.

QZ-IPCA minus IPCA; 618 paired unit-gross months. 95% pointwise percentile intervals, 12-month moving blocks, 10,000 draws. Separately recentered mean test. No adjustment for choosing a metric.

Rolling 60: full paired inference

Which parts of the performance difference are precisely estimated?

Difference	Estimate	95% interval
Rolling 60		
Annualized gross Sharpe	0.37	[0.02, 0.68]
Annualized mean (percentage points)	0.95	[−0.38, 2.36]
Annualized volatility (percentage points)	−1.29	[−2.62, 0.04]
Recentered mean-test $p = 0.17$		

The 12-month-block Sharpe interval excludes zero; longer-block checks qualify it.

QZ-IPCA minus IPCA; 618 paired unit-gross months. 95% pointwise percentile intervals, 12-month moving blocks, 10,000 draws. Separately recentered mean test. No adjustment for choosing a metric.

Changing the common sample changes the inference

What happens on the shorter sample used by the complete portfolio rule?

Metric	Δ Sharpe	95% interval	Δ mean (bp)	95% interval
EWMA	0.38	[0.09, 0.64]	6.19	[-4.12, 17.68]
Equal 120	0.34	[0.08, 0.57]	5.57	[-4.41, 16.48]
Rolling 60	0.44	[0.11, 0.75]	7.18	[-3.82, 19.76]

All three Sharpe intervals exclude zero here; the mean-return intervals do not.

594 paired months, June 1975–November 2024. Mean differences: basis points per month. 12-month blocks, 10,000 draws; pointwise intervals. The longer-block metric check uses 618 months, not this sample.

Dependence sensitivity differs across comparisons

Do confidence intervals depend on the resampling block length?

Comparison	12 months	24 months	60 months
EWMA minus IPCA	$[-0.019, 0.577]$	$[-0.048, 0.591]$	$[-0.035, 0.639]$
Equal 120 minus IPCA	$[-0.060, 0.512]$	$[-0.089, 0.533]$	$[-0.061, 0.573]$
Rolling 60 minus IPCA	$[0.019, 0.682]$	$[-0.012, 0.693]$	$[-0.010, 0.757]$
Mean-only minus off	$[1.644, 2.679]$	$[1.582, 2.840]$	$[1.421, 3.183]$
Complete minus mean-only	$[0.010, 0.035]$	$[0.011, 0.036]$	$[0.012, 0.035]$

The large mean-input Sharpe increment persists; the metric ranking is less robust.

95% paired Sharpe-difference intervals, 10,000 draws. First three rows: 618 months, QZ-IPCA minus IPCA. Last two: 594 months, one NA-IPCA fit. Equal-60's interval crosses zero at 24 and 60 months. Pointwise inference.

Forecast ranking is not portfolio ranking

Do cross-sectional prediction metrics explain the investment ordering?

Forecast	IC	Rank IC	Predictive R^2
FF5	0.0159	0.0170	-0.0004
IPCA	0.0254	0.0249	-0.0188
QZ-IPCA, rolling 60	0.0254	0.0249	-0.0188
RP-PCA	0.0364	0.0415	-0.2568
NA-IPCA, intercept off	-0.0021	-0.0046	-0.0174

IPCA and QZ-IPCA forecast metrics agree; portfolio inputs can still differ.

594 months, 1,210,039 stock-months. IC: monthly Pearson correlation; rank IC: Spearman. Predictive $R^2 = 1 - \text{SSE} / \sum r^2$ uses a zero-return benchmark. QZ-IPCA: rolling 60. The NA-IPCA row omits the intercept; it does not evaluate the complete mean forecast.

Same-fit input ablation: complete performance

Which investment input accounts for the gross-return improvement?

Panel A: Gross performance			
Statistic	Off	Mean-only	Complete
Annualized mean (%)	1.75	7.59	7.60
Annualized volatility (%)	4.46	3.02	3.00
Sharpe ratio	0.3917	2.5126	2.5347
Maximum drawdown (%)	-17.73	-3.93	-3.87

Restoring the mean changes performance substantially; instability risk changes it little.

One EWMA NA-IPCA fit, 594 months, unit gross. Complete adds a trailing-loading instability proxy with multiplier one. Off omits the intercept from investment inputs of the same fit; it is not a refitted QZ-IPCA.

Same-fit input ablation: all paired increments

How large and how precise is each incremental input effect?

Panel B: Incremental input effects			
Comparison	Monthly mean (bp)	95% interval	<i>p</i>
Mean-only minus off	48.69	[34.66, 63.36]	< 0.001
Complete minus mean-only	0.08	[-0.15, 0.26]	0.437

Comparison	Sharpe difference	95% interval
Mean-only minus off	2.1209	[1.6445, 2.6791]
Complete minus mean-only	0.0222	[0.0100, 0.0355]

The large mean contribution is distinct from the small additional Sharpe adjustment.

Mean differences: basis points per month. Sharpe differences: annualized. 95% paired 12-month-block intervals, 10,000 draws; recentered two-sided mean tests. Pointwise inference, without specification-selection adjustment.

Full sample: the increment across stock groups

Does the complete-rule increment extend beyond the aggregate Core portfolio?

Universe	Mean	95% interval	p	Holm p
Full strict-24 sample: June 1975–November 2024 (594 months)				
Core	0.488	[0.348, 0.634]	< 0.001	< 0.001
Mega	0.239	[0.105, 0.409]	0.003	0.003
Large	0.347	[0.194, 0.508]	< 0.001	< 0.001
Small	0.666	[0.487, 0.840]	< 0.001	< 0.001

Complete-minus-off mean increments are positive across all four groups.

Means and intervals: percent per month; unit gross, gross of costs. One Core NA-IPCA fit is restricted to each stock group and portfolios are re-solved. 12-month blocks, 10,000 draws. Holm adjustment covers four groups within each panel only.

Recent sample: wider uncertainty across groups

Does the complete-rule increment extend beyond the aggregate Core portfolio?

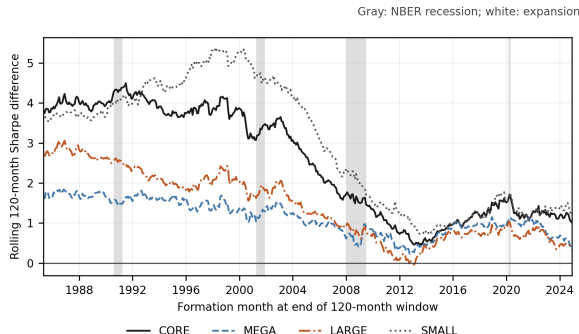
Universe	Mean	95% interval	p	Holm p
Recent period: October 2007–November 2024 (206 months)				
Core	0.202	[−0.071, 0.409]	0.11	0.32
Mega	0.206	[−0.035, 0.507]	0.14	0.32
Large	0.174	[−0.120, 0.389]	0.19	0.32
Small	0.251	[−0.061, 0.464]	0.08	0.32

Positive recent estimates coexist with intervals that include zero in every group.

Means and intervals: percent per month; unit gross, gross of costs. One Core NA-IPCA fit is restricted to each stock group and portfolios are re-solved. 12-month blocks, 10,000 draws. Holm adjustment covers four groups within each panel only.

Rolling performance shows when the increment occurs

How does complete-minus-off Sharpe vary over overlapping ten-year windows?



Core's increment is positive in every reported window, but its magnitude varies.

475 overlapping 120-month windows from the 594-month sample. Axis: final formation month in each window; gray: NBER recessions. These are not independent tests. A rolling Sharpe path and a recent mean-return interval answer different questions.

Changing the instability multiplier within a fixed fit

Is the gross result primarily generated by the extra risk adjustment?

Portfolio input	Trailing proxy	Full-rank sensitivity
Intercept off, no instability term	0.32	0.32
Intercept in mean, $\kappa_U = 0$	2.4686	2.4686
Intercept in mean, $\kappa_U = 0.25$	2.4755	2.4705
Intercept in mean, $\kappa_U = 0.5$	2.4818	2.4723
Intercept in mean, $\kappa_U = 1$	2.4926	2.4758
Intercept in mean, $\kappa_U = 2$	2.5096	2.4825

The intercept mean accounts for the large change; multiplier changes are modest.

618 months, with up to 24 prior loading estimates. κ_U multiplies the extra intercept-risk matrix. Trailing proxy and full-rank residual sensitivity matrix are distinct inputs, not validated estimator covariances. No sampling intervals; zero-multiplier equality is by construction.

Complete strategy comparison on common dates

Where do the investment rules sit beside production benchmarks?

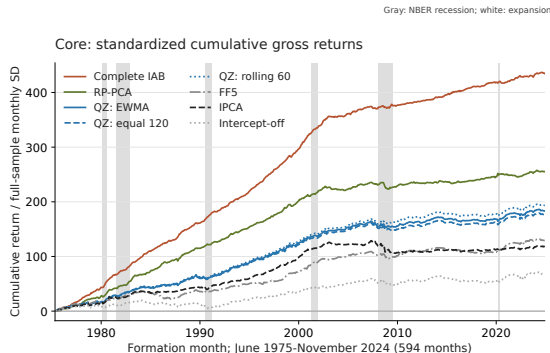
Specification	Sharpe	Volatility (%)	Drawdown (%)
FF5	0.75	5.2	−20.7
RP-PCA	1.49	3.9	−13.4
IPCA	0.69	5.9	−34.6
QZ-IPCA, EWMA	1.07	4.5	−14.2
QZ-IPCA, equal 120	1.03	4.6	−17.3
QZ-IPCA, rolling 60	1.13	4.3	−11.1
NA-IPCA, complete IAB	2.53	3.0	−3.9
NA-IPCA, intercept-off ablation	0.39	4.5	−17.7

The complete rule has the highest recorded gross Sharpe; the ablation explains its source.

594 common months; unit gross, not equal volatility; gross of costs. FF5: Fama–French five-factor model. RP-PCA: Risk-Premium PCA. IAB: intercept-aware portfolio. Full specifications, except the same-fit off control, do not isolate the metric alone.

A descriptive path comparison, not a wealth curve

When do the full strategies accumulate standardized returns?



The paths put performance timing beside the full-sample summary table.

594 months. Each path sums unit-gross returns divided by its full-sample monthly SD: $S_m(\tau) = \sum_{t \leq \tau} r_{m,t+1}^u / \hat{\sigma}_m$. Not compounding or a feasible volatility-targeting policy. Scaling is retrospective. Gray: NBER recessions.

Gross scaling is an economically consequential policy

Do raw solves and unit-gross portfolios describe the same investment?

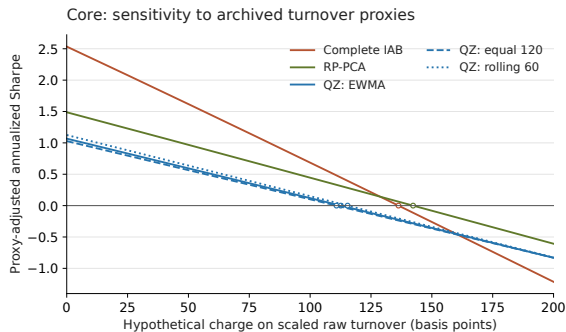
Specification	Mean gross	Raw Sharpe	Unit-gross Sharpe
FF5	1.98	0.77	0.75
RP-PCA	0.53	1.53	1.49
IPCA	3.79	0.49	0.69
QZ-IPCA, rolling mean	2.37	1.00	1.13
QZ-IPCA, EWMA	2.37	0.97	1.07
NA-IPCA, complete IAB	501	2.48	2.53
NA-IPCA, intercept-off ablation	2.27	0.38	0.39

Time-varying gross normalization changes performance; the holding policy matters.

594 months. Mean gross: average total absolute raw holdings, not mean return; raw scale follows the archived solve convention. The complete raw series includes returns below -100% and is not a feasible wealth path. The main paper evaluates the unit-gross policy.

Trading-cost sensitivity under the recorded proxy

How does an assumed one-way charge change the performance ordering?

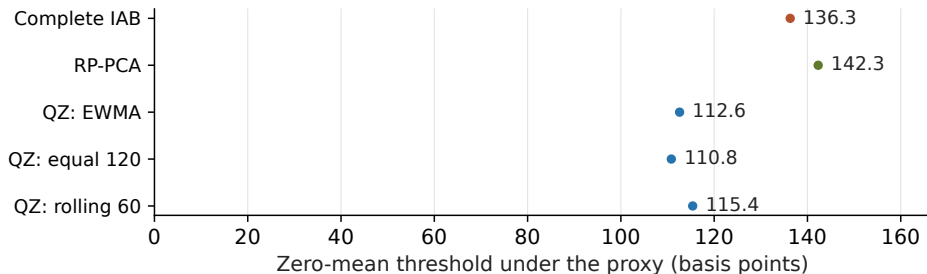


Gross Sharpe rankings need not survive increasing implementation charges.

Original panel A; 594 months. Subtract $c q_t / 10^4$ from the unit-gross return and recompute Sharpe, with c cost in basis points and q_t saved raw turnover divided by current raw gross. This is not verified turnover of drifted unit-gross holdings. Cost-proxy sensitivity, not a capacity estimate.

Gross Sharpe and cost tolerance need not rank alike

At what one-way charge does the proxy-adjusted sample mean reach zero?



RP-PCA's threshold exceeds the complete rule's, despite the latter's higher gross Sharpe.

Original panel B. Thresholds are basis points of the recorded turnover proxy (complete: 136.3; RP-PCA: 142.3). Zero sample mean is not a capacity or arbitrage bound. Different turnover exposure changes the economic comparison.

The intercept connects the model to a pricing question

What does the No-Arbitrage IPCA name motivate?

With zero conditional-mean disturbances, $\alpha_{i,t} = z'_{i,t} \Gamma_\alpha$ is conditional mean return not assigned to the fitted factor-mean terms. The budget

$$\Gamma'_\alpha Q \Gamma_\alpha \leq \delta_\alpha$$

limits training-weighted average squared stock intercepts under a Gram metric.

Using this mean in holdings is an investment-input question. Interpreting it as mispricing requires a specified economic benchmark.

Conditional pricing models can generate model-relative intercepts through omitted factors or misspecified dynamics (Zaffaroni, 2025).

A controlled retained mean makes its portfolio contribution separately measurable.

This interpretation uses predetermined characteristics and fixed coefficients within the window. A strictly positive stochastic discount factor M prices excess returns through $\mathbb{E}(M_{t+1} r_{i,t+1} \mid \mathcal{I}_t) = 0$, with $\mathcal{I}_t$ date- t information. The Gram budget does not impose this pricing moment. All saved budgets are slack.

Selected references

Which frameworks does the empirical analysis build on?

Kelly, B., S. Pruitt, and Y. Su (2019). Characteristics Are Covariances: A Unified Model of Risk and Return. *Journal of Financial Economics* 134, 501–524.

Lettau, M., and M. Pelger (2020). Estimating Latent Asset-Pricing Factors. *Journal of Econometrics* 218, 1–31.

Zaffaroni, P. (2025). Factor Models for Conditional Asset Pricing. *Journal of Political Economy* 133, 2615–2642.

Liu, M. (2026). Characteristic-Space Metrics in Factor Models: Identification and Inference. Unpublished working paper.

NBER. U.S. Business Cycle Expansions and Contractions. [Recession chronology](#).

The contribution examines investment inputs built from characteristic-based estimates.

Source: Characteristic Geometry and Portfolio Choice, V8 paper and frozen evidence. Slides reproduce reported results rather than re-estimating the raw panel.